

LECTURE 11

CLOUD & MULTIMEDIA TECHNOLOGIES IN EDUCATION



LECTURE OUTLINE



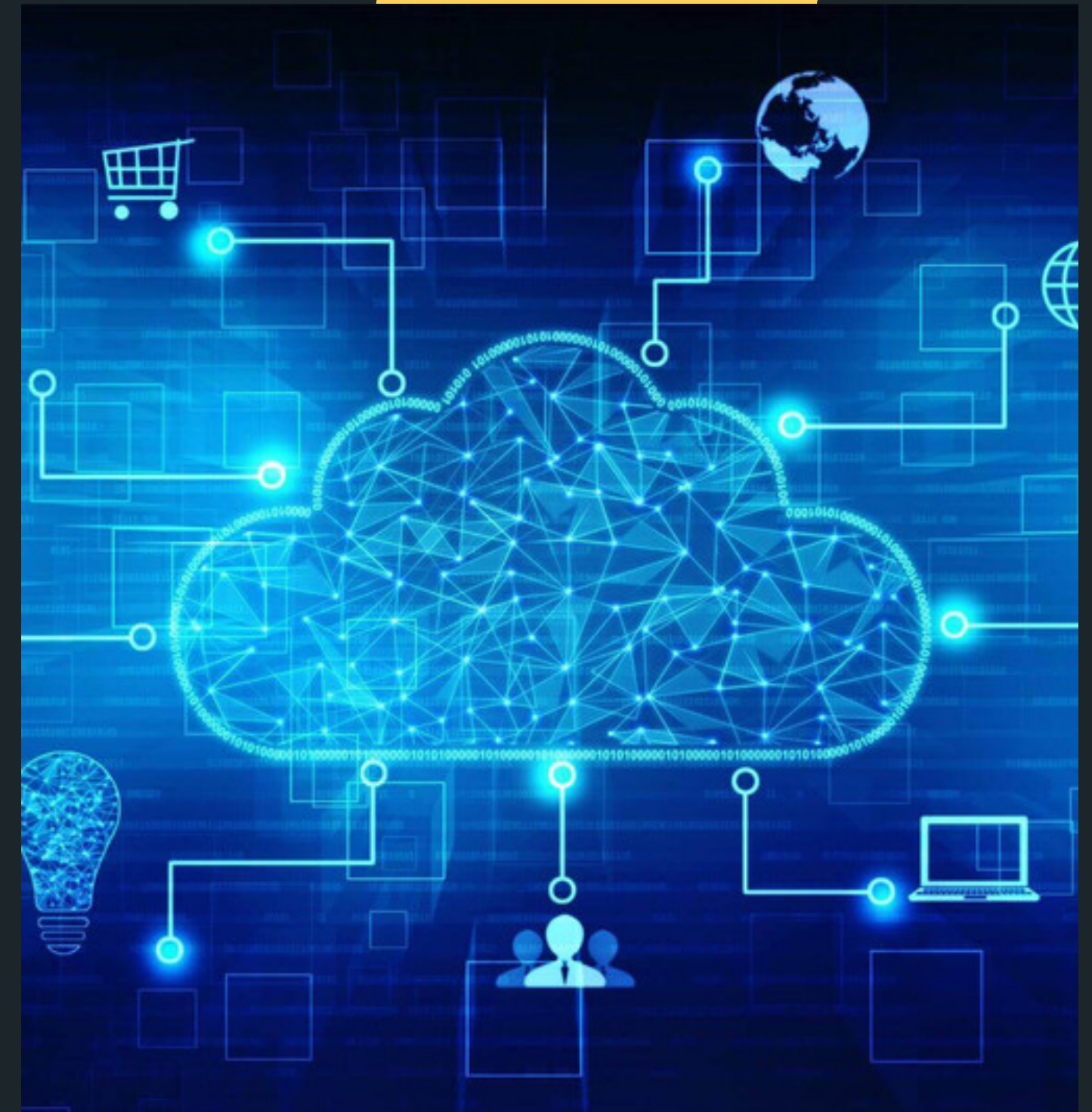
- What is cloud storage?
- How Cloud Technologies work?
- Features of Cloud technologies
- Cloud deployment models
- Cloud technology services models
- Cloud storage types
- What is multimedia technology?
- Classification of multimedia technologies
- Multimedia and its application in different areas
- Hardware and software multimedia technology
- Typer of multimedia

CLOUD STORAGE

Cloud storage is defined as a data deposit model in which digital information such as documents, photos, videos and other forms of media are stored on virtual or cloud servers hosted by third parties.

Examples of cloud services are:

- mail: gmail, hotmail;
- remote work with documents: Google documents, Office Web Apps;
- data storage: Google Drive, OneDrive, Dropbox;
- editing images in real time: Canva, Figma;
- services for creating notes, collaborating on tasks: Trello, Jira, Evernote;
- online app stores: Google Play, App Store and Microsoft Store.



Advantages of Cloud Technologies

SCALABILITY

This allows you to withstand increasing loads without disruption and data loss.

COST OPTIMIZATION

Payment for the rental of the service is made as it is used, that is, it depends on the size of the involved data volume.

SIMPLE CONTROLS

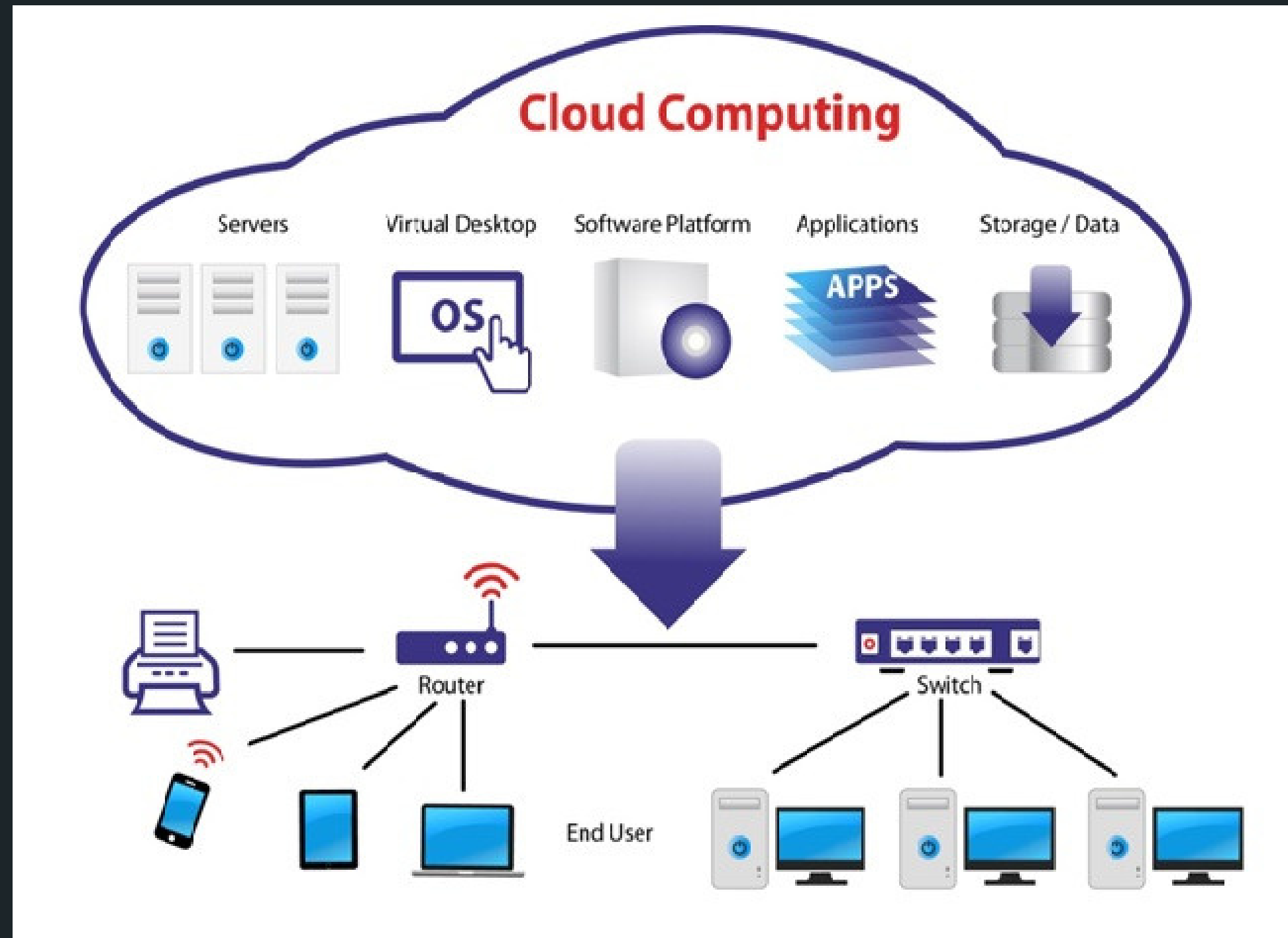
The user can independently manage the required resources by increasing or decreasing the amount of storage.

ELASTICITY

The cloud service helps to build up infrastructure without using new equipment and additional software.

The cloud is rapidly becoming the storage environment of choice for enterprises. 30% of all corporate data was stored on the cloud in 2015, which increased to 50% in 2020. The cloud storage market has also grown in tandem and is expected to be worth \$137.3 billion by 2025, as per MarketsandMarkets. This is because the cloud offers several benefits over traditional on-premise storage systems.

How Cloud Technologies work?

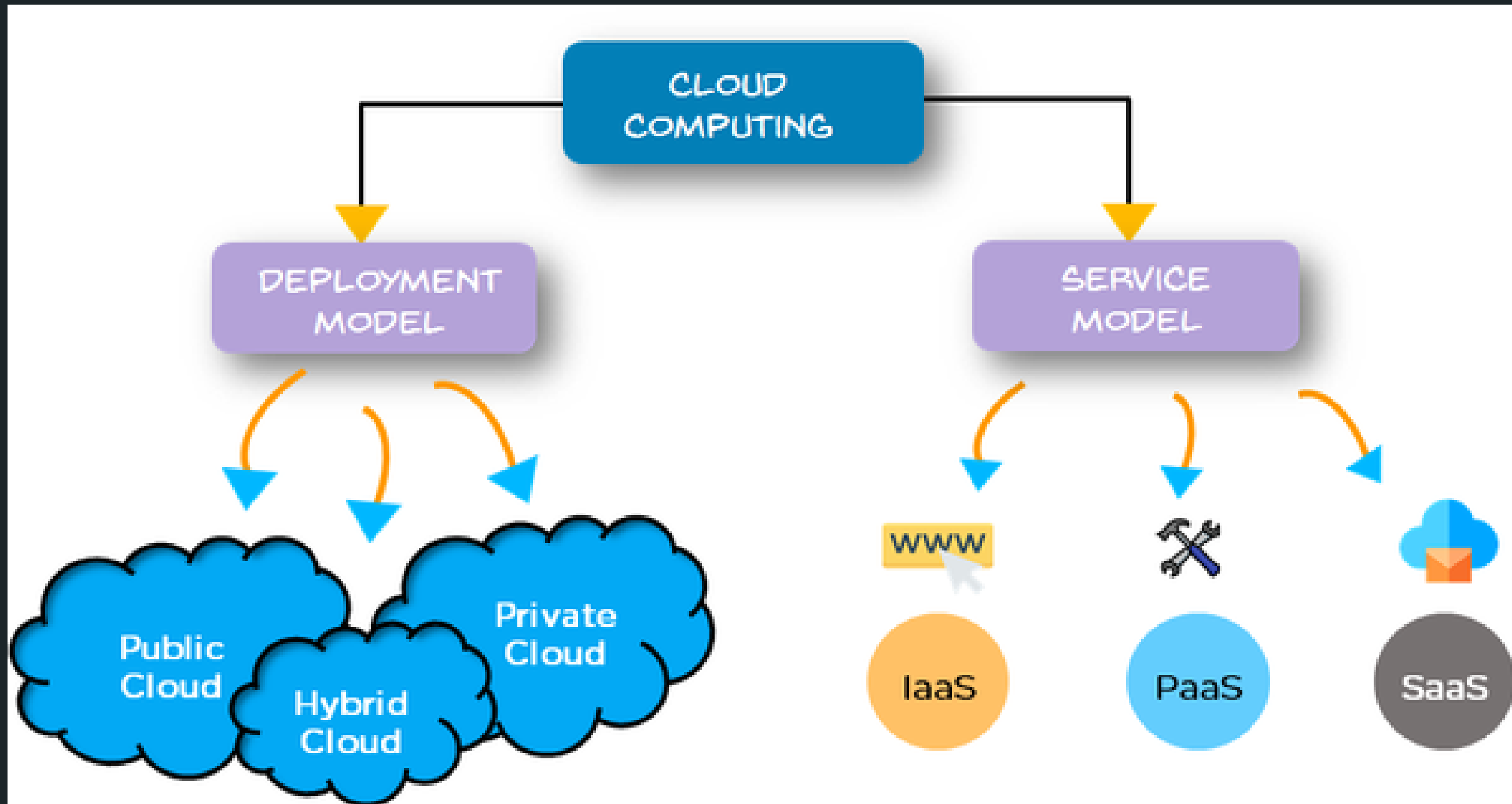


FEATURES OF CLOUD TECHNOLOGIES



- the ability to work with personal accounts and data availability from any device;
- no need to save information to a USB flash drive or other drive;
- multiple users can simultaneously edit documents and files;
- cloud services work in a browser, so it doesn't matter what operating system is on your phone, tablet or computer;
- information is stored on a cloud server - even if your PC or phone breaks down, you will not lose data;
- you are using the most recent version of the program: the service provider itself monitors its updates;
- you can share information remotely without sending a large amount of data, for example, provide access to a folder with documents or photos;

Cloud deployment models

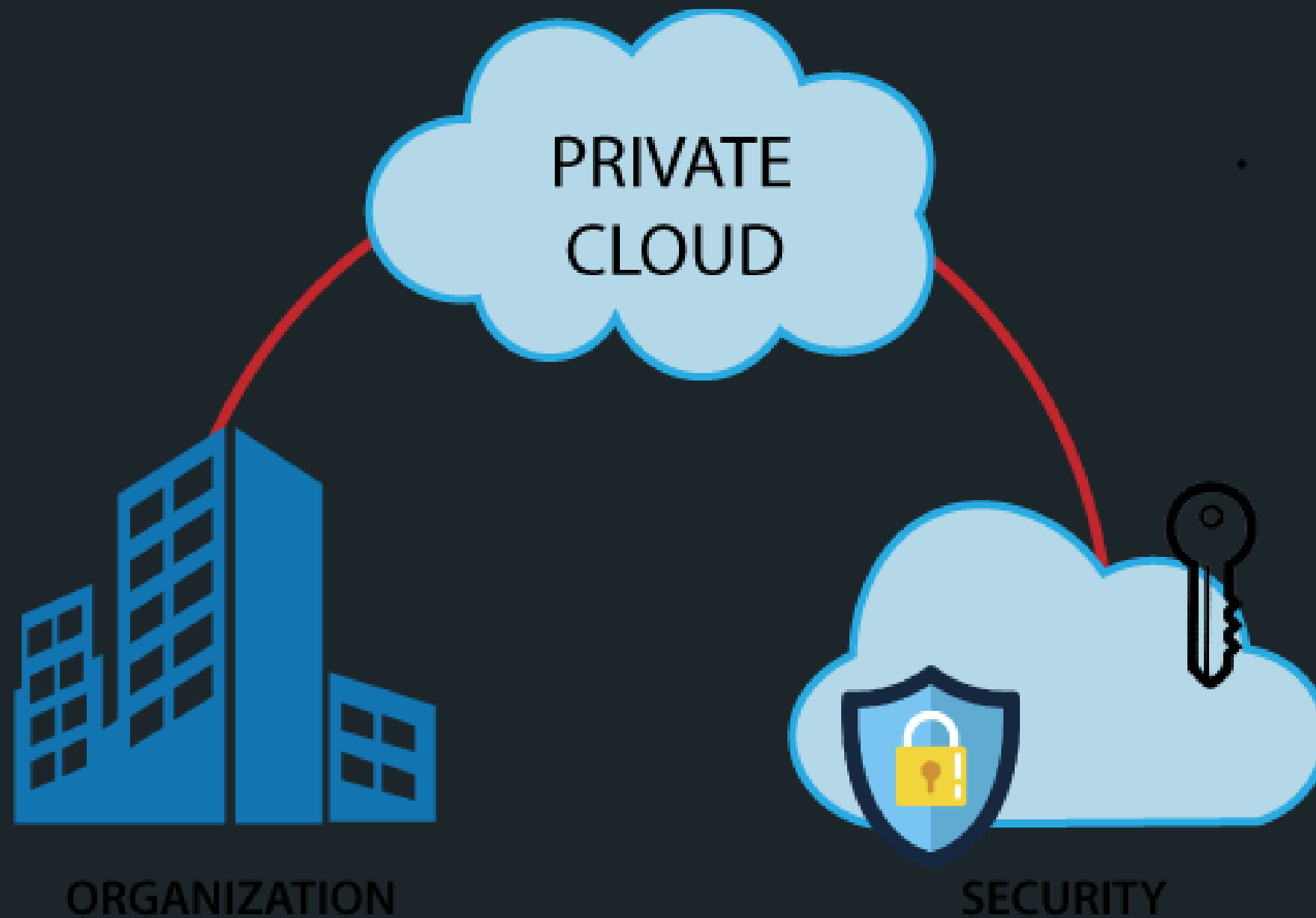


Public Cloud



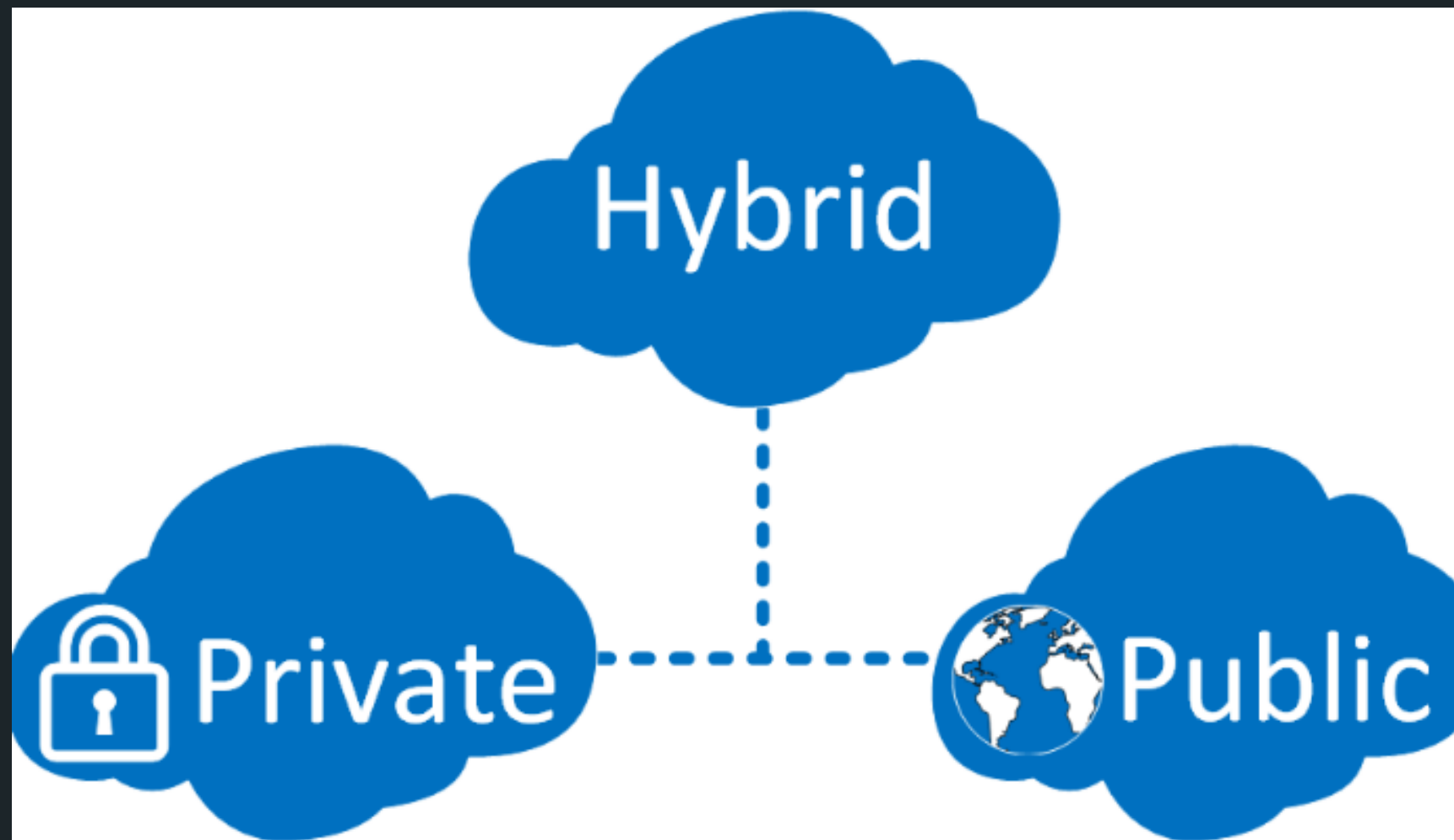
Amazon Web Services (AWS), IBM Cloud, Google Cloud, and Microsoft Azure. These are all examples of public cloud providers. Practically, these providers own and manage all the cloud architecture – hardware, software, and other infrastructure – and deliver it as a cloud service over the internet.

Private Cloud



Unlike the public cloud, the private cloud is used by only one organization. Larger enterprises with mission-critical data requiring additional security (think government agencies, financial institutions, or healthcare organizations) are more likely to use a private cloud solution. The private cloud infrastructure can reside either on-premise or with a third-party vendor. Regardless, the hardware and services are dedicated.

Hybrid Cloud



Like the name suggests, a hybrid cloud converges the public and private cloud solutions into a unique type of cloud storage that offers the advantages of both. Data and applications can move between public and private clouds as needed, offering increased flexibility. Most businesses can use the public cloud for certain aspects of their work – such as webmail – but might need a private cloud for storing secure data logs.

Cloud technology services models

Platform type	Common examples
SaaS	Google Apps, Dropbox, Salesforce, Cisco WebEx, Concur, GoToMeeting
PaaS	AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, OpenShift
IaaS	DigitalOcean, Linode, Rackspace, Amazon Web Services (AWS), Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE)

SAAS: SOFTWARE AS A SERVICE

SaaS characteristics:

- Controlled from the center
- Hosted on a remote server
- Available over the Internet
- Users are not responsible for hardware or software updates



PAAS: PLATFORM AS A SERVICE

PaaS provides cloud-based components for certain software, but is mostly used for applications.

PaaS benefits:

- Simple and cost-effective application development and deployment
- Scalability
- High availability
- Business policy automation
- Easy transition to hybrid model

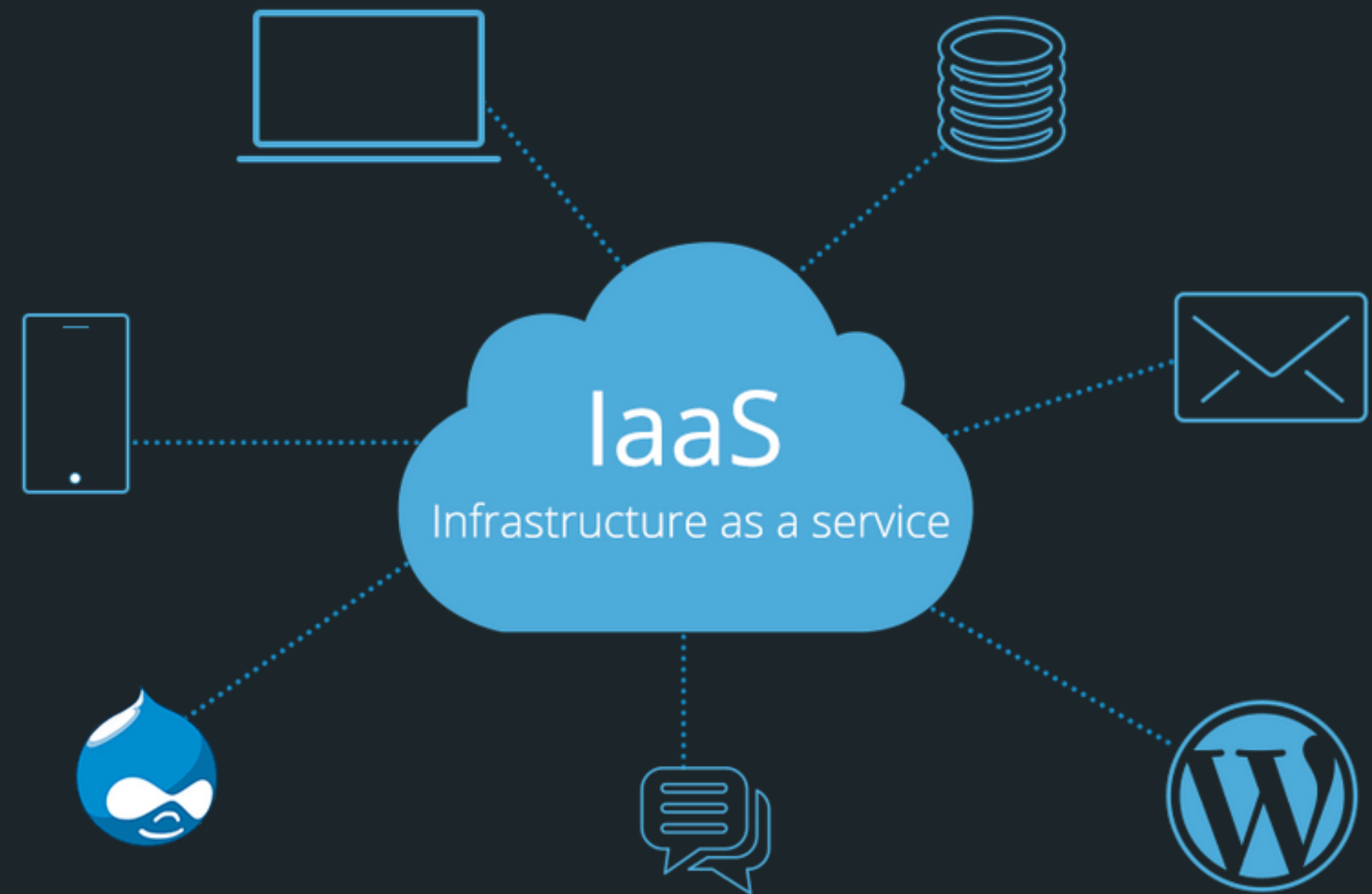


IAAS: INFRASTRUCTURE AS A SERVICE

IaaS consists of highly scalable and automated computing resources.

IaaS benefits:

- The most flexible cloud computing model
- Easily automate storage, network, server and computing power deployments
- Hardware purchase can be based on consumption
- Customers retain complete control over their infrastructure
- Resources can be purchased as needed



Cloud storage types

GOOGLE DRIVE

Google Drive allows users to store their data on servers in the cloud and share it with other users on the Internet. Cloud storage divides the space between Google Drive, Gmail, and Google Photo. The service can store not only documents, but also photos, music, videos and many other files - 30 types in total. Everything is very convenient and familiar for users of Google services.



GOOGLE DRIVE

Memory Space

15 GB

100 GB

200 GB

2 TB

Monthly Fee

Free

690 KZT

990 KZT

3390 KZT

The maximum file size is 5 TB.

Available on web browsers,
Windows, Mac OS, Android, iOS,
and more.

Google Drive website:
<https://drive.google.com>

ONE DRIVE

OneDrive is a cloud-based online file storage service with file sharing functions. The advantage of OneDrive is that it is immediately integrated with Office 365, so you can create, edit, save Excel, OneNote, PowerPoint, and Word files directly from the app to Windows Live OneDrive.

The OneDrive service allows you to store at the moment 5 GB of information for free in a format organized using standard folders.

Microsoft OneDrive website:

<http://onedrive.com>



DROPBOX

Dropbox is a cloud storage solution that allows users to store their data on servers in the cloud and share it with other users on the Internet. Its work is based on data synchronization. Dropbox features:

- 256-bit AES encryption and SSL encryption;
- Best-in-class sync technology;
- Integration with Microsoft Office 365;
- Unlimited file recovery and version history;
- Access links with password and expiration date;
- Customizable access levels, etc.

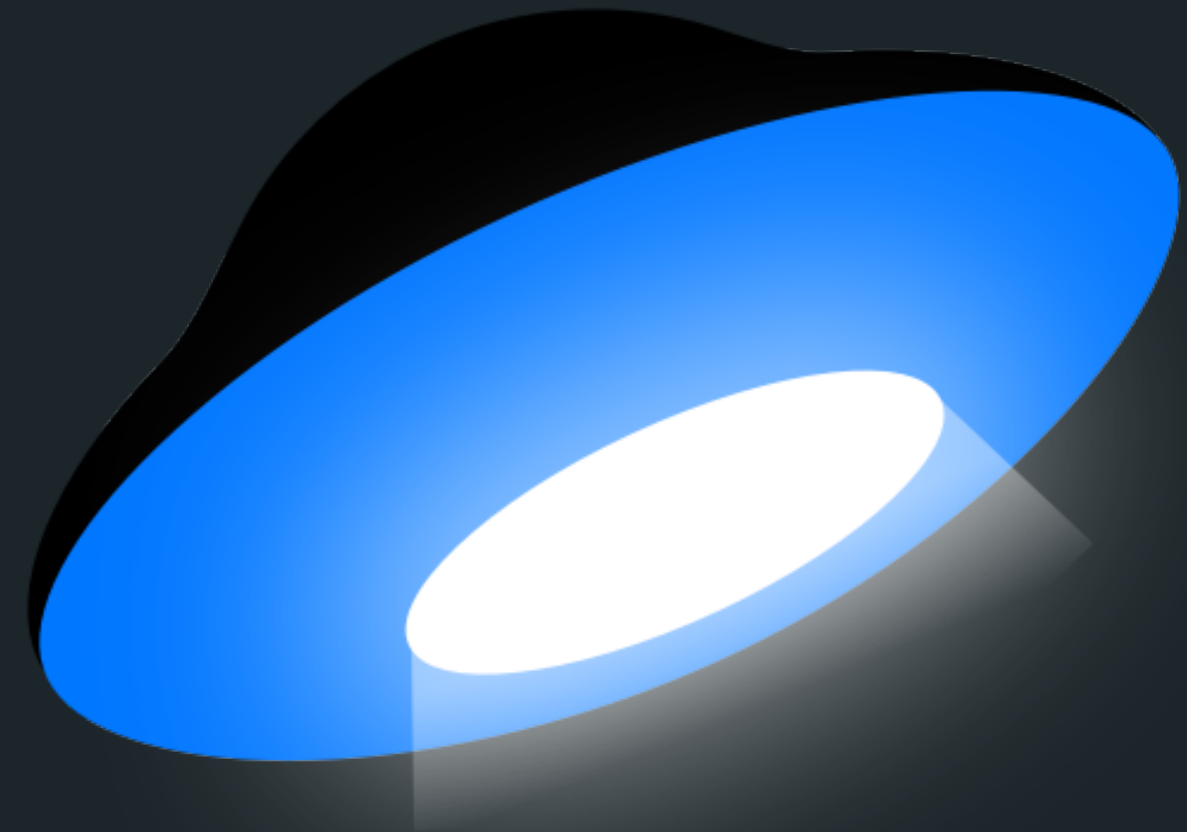
Dropbox website: <http://www.dropbox.com>



YANDEX DISK

Yandex Disk is a Russian cloud service that allows users to store their data on servers in the cloud and transfer it to other users on the Internet. 10 GB cloud space is free for everyone. Yandex Disk can act as a cloud service, integrating into the Microsoft Office suite.

Yandex.Disk website: <http://disk.yandex.ru/>



AMAZON WEB SERVICES (AWS)

Amazon Web Services (AWS) is not just a cloud-based data warehouse, but a collection of global services. Thus, Amazon Web Services includes a wide range of fundamental cloud infrastructure services:

- Data storage and content delivery
- Computing
- Databases
- Networking solutions

AWS has a rich selection of services to improve the efficiency of your cloud: analytics, enterprise applications, mobile services, IoT.

Amazon Web Services website:

<https://aws.amazon.com>



ICLOUD

iCloud Drive is Apple's cloud storage. If you have an Apple device, then you already have iCloud Drive installed on your device. You just have to set up an iCloud account. For data security, iCloud Drive encrypts your data using 128-bit AES encryption. It also uses 128-bit SSL encryption for transmission. 5 GB space is free upon registration.

iCloud Drive website:

<https://support.apple.com/en-us/HT201318>



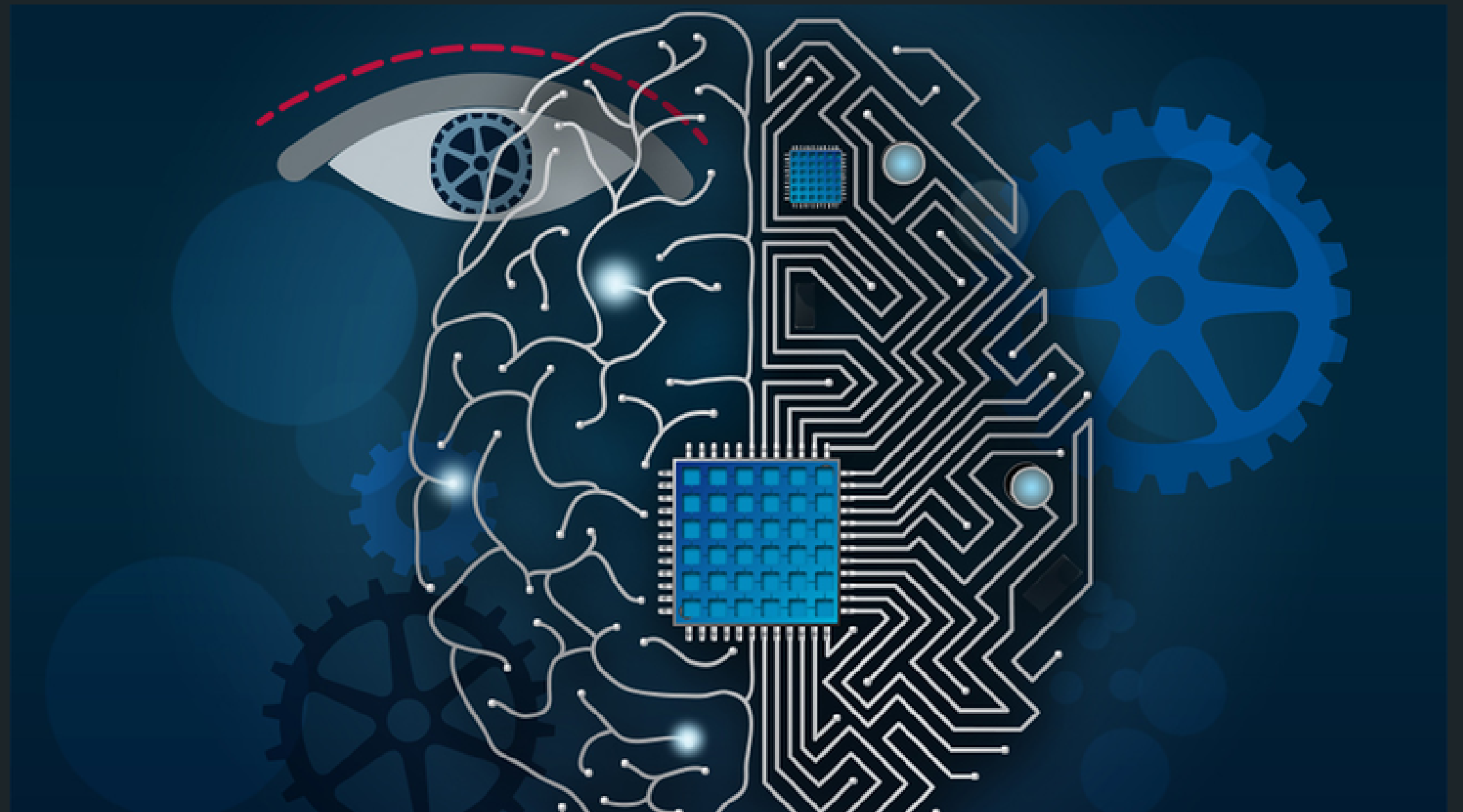


Multimedia technologies

MULTIMEDIA TECHNOLOGY

Multimedia is a set of hardware and software tools that allow the user to work in an interactive mode with heterogeneous data (graphics, text, sound, video and animation), organized as a single information environment.





The use of multimedia ensures the ease of perception of information by a person, since a person has means and methods of information processing that are significantly different from a computer and have a form of perception that is convenient for a person.

Multimedia

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graph TD; A[Multimedia] --> B["- based on support for interoperability<br/>- based on the use of various multimedia<br/>telecommunication technologies"]; A --> C["- Linear. The analogue of the linear<br/>presentation method is movie. The person<br/>viewing this document cannot in any way<br/>influence its conclusion.<br/>- Non-linear. A non-linear way of presenting<br/>information allows a person to participate in<br/>the display of information by interacting in<br/>some way with the media display."];
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Multimedia and its application





EDUCATION

Entertaining education;
Computer-based training
courses;
Discrete technologies, etc.



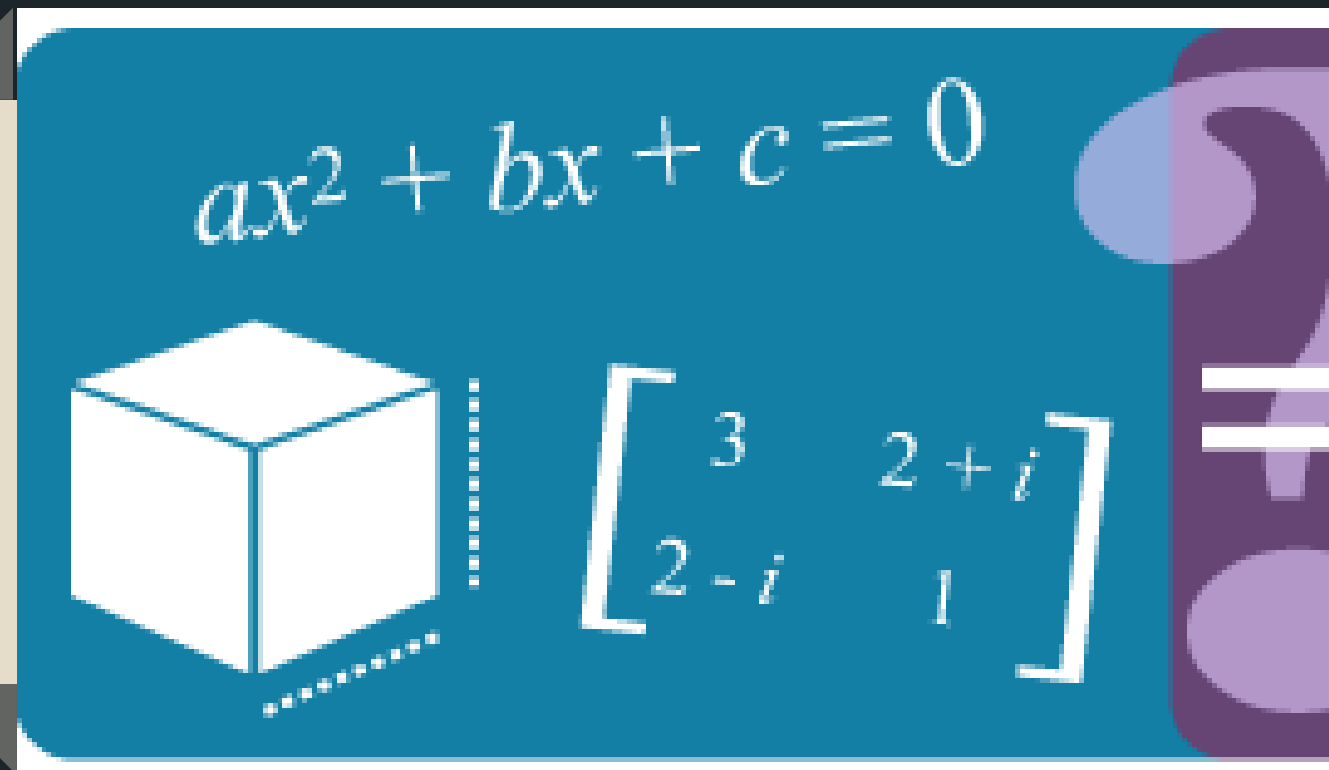
SOFTWARE DEVELOPMENT

Interface media;
Computer simulations;
User-friendly software, etc.



INDUSTRY

to present information to
shareholders;
useful in organizing staff
training, advertising, and
product sales



MATHEMATICAL AND SCIENTIFIC RESEARCH

multimedia is mainly used for modeling and simulation

MEDICINE

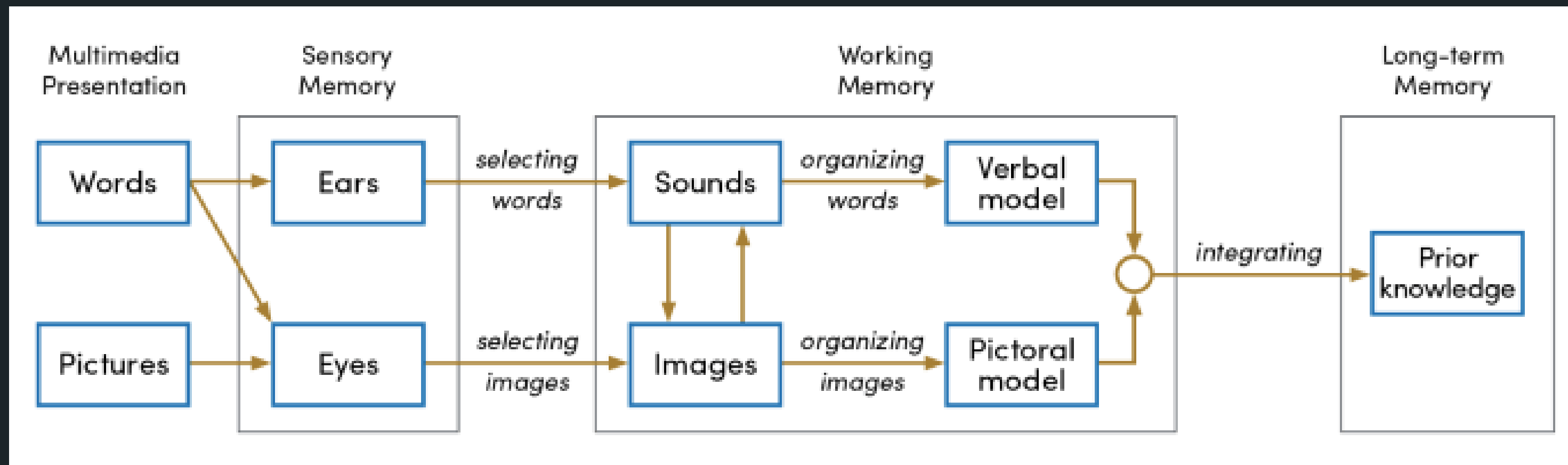
virtual surgeries or simulations of a human body



COGNITIVE THEORY OF MULTIMEDIA LEARNING

Multimedia instructional content is more likely to create a meaningful learning experience if the content is developed with the following assumptions from cognitive science in mind:

- Active processes assumption: Active learning entails carrying out a coordinated set of cognitive processes during learning.
- Dual-channel assumption: Dual channels, one for visual/pictorial and one for auditory/verbal processing.
- Limited-capacity assumption: Each channel has limited capacity for processes.



Types of multimedia

- ***Interactive Multimedia***

The first is interactive multimedia, a form of multimedia that the user can fully control what and when the multimedia is displayed or run. The easiest examples of this multimedia are games and virtual reality. This type of multimedia is arguably the most interesting and enjoyable because we can interact more with it to better feel the content.

- ***Hyperactive Multimedia***

This type of multimedia has a structure of various related elements that the user can control. Although it is not like interactive multimedia, this type of hyperactive multimedia also has certain functions. For example, websites and online games.

Types of multimedia

- ***Linear/Sequential Multimedia***

Then there is the type of linear multimedia or sequential multimedia. Linear multimedia is a type of multimedia that runs linearly or in a straight line. This type of multimedia only moves in the same direction as we often encounter in all types of video tutorial films.

In contrast to interactive multimedia, there is an interaction between the user and the multimedia by using computers, mice, keyboards. At the same time, linear multimedia in its use must be scheduled and sequentially from start to finish. Examples such as music and television station broadcasts.

Types of multimedia

- ***Multimedia Learning Presentation***

The next type of multimedia is learning presentation multimedia. Especially in the current pandemic era where all activities are limited, including one of them in school, a teacher must continue to teach online.

The existence of learning presentation media like this will greatly facilitate a teacher in delivering material online. Therefore indeed, multimedia also has an important function like this, which has a lot of positive impacts on humans.

- ***Interactive Media***

The next type of multimedia is interactive media, where interactive media is a learning system that has previously been recorded, whether visual, sound, or even video, and presented under the control of a computer. By using interactive media, users will see and hear the sound of the learning and make active responses so that the learning process can be more interesting.

Types of multimedia

- ***Self-Learning Multimedia***

Then there is home self-learning multimedia, where multimedia like this is multimedia in the form of software and can be used by students to independently study various materials that have been prepared in the independent learning multimedia and do not need teacher assistance.

In the current pandemic era, self-learning multimedia is also one thing that will be very useful, considering that offline teaching and learning activities in schools are still not permitted, so software like this will be very helpful. Independent learning multimedia combines knowledge and tacit knowledge, for example, Adobe Flash.

- ***Virtual Reality***

Maybe this is one type of multimedia that is currently very popular, namely virtual reality. One advantage of virtual reality is that you can feel the simulation that you are in a certain place or are doing something real through visual displays and sound.